Abstract

This project presents a smart umbrella system integrating GPS tracking, acceleration-based theft detection, 4G communication, and an alert mechanism. Through a step-by-step development process, we explored different positioning solutions—Bluetooth+GPS, WiFi+GPS, and ultimately 4G+GPS—evaluating each for feasibility and reliability. The final design enables real-time umbrella location tracking via cloud server, while anti-theft alerts are locally triggered through onboard acceleration sensing and a button-controlled activation mechanism. The system is built on Arduino Uno with peripheral modules such as GT-U8 GPS, MPU6050 accelerometer, and Air780EX 4G module. A 3D-printed encapsulation ensures practical usability. This report documents the design, implementation, and testing of the system, offering insights into IoT-based product development.

1 Introduction

1.1 Background

In the era of rapid - paced technological advancement, the integration of smart features into everyday items has become a significant trend. Umbrellas, as a common daily necessity, have long remained in a relatively traditional form. However, with the increasing demand for convenience, functionality, and security in people's lives, there is a growing need to transform traditional umbrellas into smart ones. The development of technologies such as the Internet of Things (IoT), sensors, and wireless communication provides a solid technical foundation for this transformation. Smart umbrellas have the potential to not only meet basic rain - protection needs but also offer a series of intelligent functions, bringing more convenience and new experiences to users.

1.2 User needs

·**Anti - loss demand**: Traditional umbrellas are often lost due to forgetfulness or theft. Users hope that smart umbrellas can have effective anti - loss measures, such as real - time location tracking and anti - theft alarm functions, so that they can quickly find their umbrellas when they are lost.

·**Function expansion demand**: In addition to the basic function of rain protection, users expect umbrellas to have more practical functions. For example, in low - light environments, lighting functions can improve the safety of travel; in special situations, the umbrella can be used as a signaling tool.

·**Personalization and information management demand**: Users hope to be able to customize and manage information related to the umbrella, such as personal contact information on the umbrella, so that when the umbrella is lost, it can be more easily returned.

1.3 Review of existing products

In the current market, several innovative umbrella designs aim to enhance user convenience and safety. Two notable examples are:

1. Umbrella with Password Lock on the Handle

This type of umbrella integrates a combination lock into the handle. The umbrella remains folded and cannot be opened unless the correct password is entered, effectively preventing unauthorized use or theft.

Advantages: Prevents unauthorized use; Enhances anti-theft functionality; Simple locking mechanism for everyday use.

Disadvantages: If the password is forgotten, the umbrella cannot be used; Mechanical parts increase the risk of failure; Adds weight and complexity to the handle

2. Umbrella with Flashlight at the Bottom of the Handle

This umbrella features a built-in flashlight at the base of the handle, offering illumination during dark or rainy conditions.

Advantages: Improves visibility in low-light environments; Adds safety when walking at night

Disadvantages: Adds complexity and potential water leakage at the junction point; Provides no additional function in daily use besides lighting; Limited lighting range and intensity

1.4 Customer needs

- **Customer needs**

·**The need to prevent umbrella loss**: A survey of 500 umbrella users shows that 78% of the respondents have lost at least one umbrella in the past year. Among them, 42% were lost in public transportation places, and 30% were lost in commercial areas such as shopping malls and coffee shops. Users are eager for smart umbrellas to have reliable anti-loss functions, such as real-time positioning and out-of-distance alarm, so that they can be quickly retrieved after being lost, reducing economic losses and inconvenience in daily life.

·**Lighting requirements in low-light environments**: In a survey on the use of umbrellas in low-light conditions, 82% of the respondents indicated that they had encountered difficulties, and 55% had nearly tripped over obstacles while walking with an umbrella at night. Users expect umbrellas to provide sufficient and practical lighting in low-light conditions, such as adjustable brightness and flashing warning modes, to ensure travel safety.

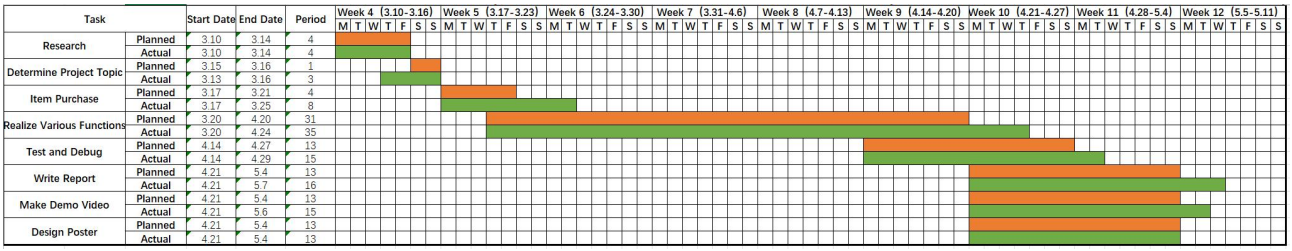
·**Durability requirements**: 90% of users are concerned about the durability of umbrellas, hoping that they can withstand bad weather such as strong winds and heavy rain. Users expect umbrellas to be more stable in structural design and more durable in material selection to ensure normal use in extreme weather.

·**Waterproof requirements**: In the research of usage scenarios in rainy environments, 68% of users were concerned that rainwater intrusion would affect the normal operation of the smart functions of umbrellas. They put forward higher requirements for the waterproof performance of the electronic components of umbrellas, expecting smart umbrellas to have good waterproof capabilities to ensure the safe operation of electronic devices in rainy days.

·**Comprehensive practical requirements**: Users hope that the umbrella, on the basis of having the function of being suitable for both sunny and rainy days, will further enhance its performance. For instance, enhance the waterproof effect of the outer waterproof coating, extend the effective duration of the inner sun protection and cooling material, and at the same time, take into account both aesthetics and portability in the design to meet the diverse needs of daily travel.

1.5 Project planning and workload distribution

The planned project work progresses is shown in Figure.



Mengxuan Shen	25%	Hardware Selection and Procurement, NFC information editing test, The iterative process of connecting and arranging the light strips, Confirmation of 3D printed model, video editing, and Report Writing
Haiqi Shen	25%	Integration of GPS and communication modules, The evolution from Bluetooth to Wi-Fi to 4G, Setup of the server, Video recording and report writing
Chenxin Li	25%	Acceleration module and NFC writing implementation, The iterative process of connecting and arranging the light strips, System structure diagram drawing, Video recording and report writing
Jiayi Peng	25%	Design concept proposal, Research on NFC Components, Connection of the light strips, poster production, Video recording and report writing

2 Design concept comparison, development and screening

- **Multiple design schemes**

1) Anti-loss design plan

·**GPS + Bluetooth combined tracking**: Equipped with a high-precision GPS module and a low-power Bluetooth module. The GPS module continuously locates the position of the umbrella. When it is lost at a long distance, the position of the umbrella is displayed in real time through the mobile phone APP using the satellite map. When the umbrella enters the effective Bluetooth connection range (within 100 meters), the Bluetooth module automatically wakes up, and the mobile phone APP can provide precise navigation for finding the item, guiding the user to locate the umbrella. At the same time, when the umbrella and the mobile phone exceed the preset Bluetooth connection distance, the mobile phone and the umbrella end will simultaneously trigger an alarm to remind the user.

·**GPS+WiFi combined tracking:** Equipped with GPS module and WiFi module. In an outdoor open environment, the GPS module works normally, accurately locating the position of the umbrella in real time and sending the location information to the user's pre-set receiving terminal, such as a personal computer or tablet computer, via the 4G communication module. When the umbrella enters an indoor area or an area covered by WiFi, the WiFi module automatically searches for and connects to the available WiFi networks around, using WiFi base station positioning technology to obtain more accurate location information (with higher positioning accuracy compared to simple GPS indoors). After obtaining the location data, the information is transmitted to the receiving terminal through the 4G communication module. In addition, users can set the information of commonly used WiFi networks such as home and

office locations on the receiving terminal. When the umbrella is connected to these preset networks, the receiving terminal will automatically pop up a prompt to inform the user of the approximate location of the umbrella.

·**GPS+4G communication combined tracking**: Integrated GPS positioning chip and 4G communication module. The GPS module acquires the real-time location information such as longitude, latitude and altitude of the umbrella. The 4G communication module sends the data to the user's pre-set receiving terminal, such as personal computer or tablet computer, through the mobile network. The user sets the receiving address and parameters on the receiving terminal. The positioning data is transmitted at a fixed frequency (such as once per minute). The receiving terminal uses the online map service to visually display the current position of the umbrella, facilitating the user to quickly lock the direction.

·**NFC recognition**: An NFC chip is embedded inside the umbrella handle. When used for the first time, users can write their personal information (such as name, contact details, home address, etc.) into the chip in encrypted NDEF format with the help of an Arduino development board and an NFC reading and writing module. In subsequent use, users can turn on a mobile phone that supports NFC function, pair it with a universal NFC editing APP, and simply hold the umbrella handle close to read and modify the information. When others pick up an umbrella, they can approach the chip with an NFC-enabled mobile phone to obtain the owner's information, which is convenient for timely return. The chip adopts data encryption and permission control technology. Only authorized apps and verified by users can modify information.

2) Abnormal behavior detection

For theft - prevention - focused abnormal behavior detection, an accelerometer is a key component. It continuously monitors the movement of the umbrella. When someone tries to steal the umbrella, actions like sudden, unauthorized movement or rapid changes in the umbrella's position will cause distinctive acceleration changes that the accelerometer can detect. Once such suspicious acceleration patterns are identified, indicating a potential theft attempt, the system is programmed to trigger the buzzer to emit a loud alarm. This draws attention and deters the thief. To prevent false alarms from the buzzer, a button is incorporated. The user can press this button to quickly disable the alarm in case of accidental triggering due to normal movements that might have been misinterpreted as theft - related actions. This way, the alarm system remains effective in detecting actual theft while minimizing disruptions from unnecessary alerts.

3) Lighting design scheme

·**Programmable light strips**: Programmable LED light strips are arranged along the edge of the umbrella surface and at the ribs. Users can make diverse Settings for the light strips through the accompanying simple controller or the mobile phone mini-program. It can not only adjust the brightness and switch colors, but also customize the flashing frequency of the lights, gradient mode, marquise-like effects, etc. For example, during daily travel, it is set to a soft white always-on mode for lighting. In special circumstances, it can be set to flash rapidly in red as a warning signal to meet the needs of users in various scenarios.

·**LED light strips**: High-brightness and low-energy consumption LED light strips are selected and installed on the edge of the umbrella surface and the ribs. Simple control can be achieved through the physical buttons on the umbrella handle. A short press of the button can turn the light strip on or off, and a long press of the button can adjust the brightness of the light strip. There are three brightness adjustment levels: low brightness (suitable for close-range lighting and power saving), medium brightness (for daily use), and high brightness (for dealing with dark, foggy and other harsh environments).

4) Durability and practicality improvement plan

·**Structural reinforcement**: The connection points of the 32 fiberglass bones were strengthened, and more secure connectors were used to enhance the overall structural stability. The junction of the 14mm thickened fiber middle bar and the ribs is optimized to enhance the bending resistance under strong wind and heavy rain.

·**Material upgrade**: The outer waterproof coating has been improved by adopting nano-level waterproof technology to enhance waterproof performance and durability. Upgrade the inner layer of sun protection and cooling materials, select more efficient heat insulation and sun protection materials, and extend the duration and effect of sun protection.

5) Waterproof electronic packaging solution

·**Multi-layer sealed protection**: All electronic components (GPS module, Bluetooth module, 4G communication module, WiFi module, lighting control circuit, etc.) are placed in a specially designed waterproof sealed cabin. The sealed cabin adopts a multi-layer structure. The inner layer is made of insulating and moisture-proof materials to isolate moisture. The middle layer is a waterproof rubber ring, forming a sealing ring around the edge of the cabin. The outer layer is made of a high-strength waterproof plastic shell, which is fastened with screws to ensure overall sealing and prevent rainwater from entering and affecting the normal operation of electronic components.

6) Bluetooth integrated entertainment functions

Incorporate a high-quality Bluetooth audio module into the umbrella. When paired with a smartphone, users can answer calls directly through a built-in microphone and speaker on the umbrella handle, making it convenient to communicate during rainy days without having to take out the phone. Additionally, it can play music, allowing users to enjoy their favorite tunes while walking. This function can be controlled via buttons on the umbrella handle, enabling operations such as play/pause, track switching, and volume adjustment.

7) Solar charging function

Install a thin-film solar panel on the surface of the umbrella. The solar panel can convert sunlight into electrical energy and store it in a built-in rechargeable battery. This stored energy can be used to power the electronic components of the umbrella, such as the GPS module, lighting system, and Bluetooth module, reducing the reliance on replaceable batteries and enhancing the product's sustainability.

8) Environmental monitoring function

Integrate sensors for measuring temperature, humidity, and air quality into the umbrella. The umbrella can collect real-time environmental data and transmit it to the user's mobile phone APP. This allows users to be aware of the surrounding environmental conditions, for example, checking the humidity level before going out or monitoring the air quality in their area.

- ## Design scheme comparison

Functionality:

Anti-loss tracking: The GPS + Bluetooth combined tracking excels in both long and short-range anti-loss, offering an alarm feature for disconnection. GPS + WiFi combined tracking provides accurate indoor positioning, complementing the outdoor GPS functionality. GPS + 4G communication combined tracking ensures stable long-range positioning, making it reliable for tracking the umbrella from afar. NFC recognition focuses on facilitating umbrella return by allowing others to access the owner's information easily.

Security: The accelerometer-based alarm is highly effective for theft prevention, promptly detecting abnormal movements and triggering an alert.

Lighting: Programmable light strips offer a high degree of customization, enabling users to adjust brightness, color, and lighting effects according to different scenarios. LED light strips, on the other hand, provide simple and practical lighting with basic on/off and brightness adjustment functions.

Durability and protection: Structural reinforcement and material upgrade enhance the umbrella's physical durability and protective capabilities against weather elements. The multi-layer sealed protection safeguards the electronic components from water damage.

Extra features: Bluetooth integrated entertainment functions add convenience for communication and entertainment. The solar charging function promotes sustainability by providing an alternative power source, while the environmental monitoring function offers useful environmental information to users.

Technical Difficulty:

GPS + Bluetooth combined tracking and GPS + 4G communication combined tracking have moderate technical challenges mainly in integrating the respective modules. GPS + WiFi combined tracking is more technically demanding due to the need to switch between GPS and WiFi and handle indoor positioning accurately. NFC recognition has a low technical difficulty level, mainly involving basic chip reading and writing operations. The accelerometer-based alarm and LED light strips are relatively simple in terms of technical implementation. Programmable light strips require more complex programming for diverse lighting effects. The multi-layer sealed protection demands sophisticated design and assembly for a reliable waterproof seal. Bluetooth integrated entertainment functions, solar charging, and environmental monitoring functions all involve high technical complexity, including module integration, software development, and data processing.

Cost:

GPS + 4G communication combined tracking incurs high costs mainly due to the 4G module. GPS + Bluetooth combined tracking and GPS + WiFi combined tracking have moderate costs considering the module combinations. NFC recognition is cost-effective with the low cost of the NFC chip. The accelerometer-based alarm and LED light strips are relatively inexpensive. Programmable light strips can be costly due to their advanced features. Structural reinforcement and material upgrade bring moderate cost increases. The multi-layer sealed protection, Bluetooth integrated entertainment functions, solar charging, and environmental monitoring functions all come with high costs due to the materials, components, and development efforts involved.

User Demand Match:

Anti-loss tracking methods like GPS + Bluetooth, GPS + WiFi, GPS + 4G, and NFC recognition all strongly align with users' core need to prevent umbrella loss. The accelerometer-based alarm meets users' concern for theft prevention. LED light strips satisfy the basic lighting needs of users, while programmable light strips have a more moderate match as lighting customization is not an essential requirement for most. Structural reinforcement, material upgrade, and multi-layer sealed protection are highly relevant to users' demand for a durable and reliable umbrella. Bluetooth integrated entertainment functions, solar charging, and environmental monitoring functions have lower user demand match as they are more supplementary features rather than primary needs for an umbrella.

- ## Function screening

Based on the above comparison, the following functions are selected for implementation:

GPS + 4G communication combined tracking: It offers stable long - range positioning, which is crucial for anti-loss, and although the cost is relatively high, it meets the core user demand for accurately tracking the umbrella's location from a distance.

NFC recognition: With low technical difficulty and cost, it effectively helps with umbrella return, which is a practical and user - friendly feature.

Accelerometer-based abnormal behavior detection: It provides reliable theft prevention with a simple structure and low cost, addressing an important user concern.

LED light strips: This option provides basic and practical lighting functions with simple control, meeting the daily lighting needs of users at a moderate cost.

Structural reinforcement and Material upgrade: These two measures significantly improve the durability and practicality of the umbrella, enhancing the overall user experience and product lifespan, and the cost and technical difficulty are within an acceptable range.

Multi-layer sealed protection: Essential for protecting the electronic components inside the umbrella, ensuring the normal operation of other functions in various weather conditions, despite the relatively high technical difficulty and cost..

● Function Overview

1) Identity recognition function

The smart umbrella is equipped with an NFC module. When used for the first time, users can write their personal information (such as name, contact details, home address, etc.) into the NFC chip inside the umbrella handle in encrypted NDEF format through the Arduino development board and the NFC reading and writing module. In subsequent use, users only need to turn on a mobile phone that supports NFC function, pair it with a universal NFC editing APP, and bring the phone close to the handle of the umbrella to easily read and modify the personal information in the chip. When someone picks up an umbrella, they can bring an NFC-enabled mobile phone close to the chip to obtain the owner's information, which is convenient for timely return. Meanwhile, the NFC chip adopts data encryption and permission control technology. Only authorized apps and verified by users can modify information to ensure information security.

2) Anti-theft alarm function

Integrate the acceleration sensor with the buzzer and install it inside the umbrella. The acceleration sensor can monitor the acceleration changes of the umbrella in real time. When the umbrella is stationary, the acceleration is at a relatively low level. Once someone picks up an umbrella without permission and moves it quickly, causing the umbrella's acceleration to exceed the preset threshold, the acceleration sensor will immediately detect this change and transmit the signal to the control system. After receiving the signal, the control system will immediately activate the buzzer and emit a high-decibel alarm sound to warn the umbrella thief and also draw the attention of people around.

3) Real-time positioning function

Users need to set the receiving address and related parameters on the receiving terminal in advance. When the umbrella is lost, the positioning data will be transmitted to the receiving terminal at a fixed frequency (such as once every ten minutes). After receiving the data, the receiving terminal visually displays the current location of the umbrella through online map services (such as Baidu Maps API, Autonavi Maps API, etc.), helping users quickly lock the position of the umbrella.

4) Lighting function

The LED light strip is integrated inside the umbrella to achieve uniform lighting through the optical fiber light guide technology, and automatically adjust the brightness combined with the switch to solve the potential safety hazard in the low light environment in rainy days. Two modes are supported:

Steady flash: continuous lighting, providing stable visible light (preventing roadblocks in the dark at night).

Alert flash: high frequency flashing (2Hz), used for emergency warning (such as obstacles approaching, abnormal tilt of the umbrella).

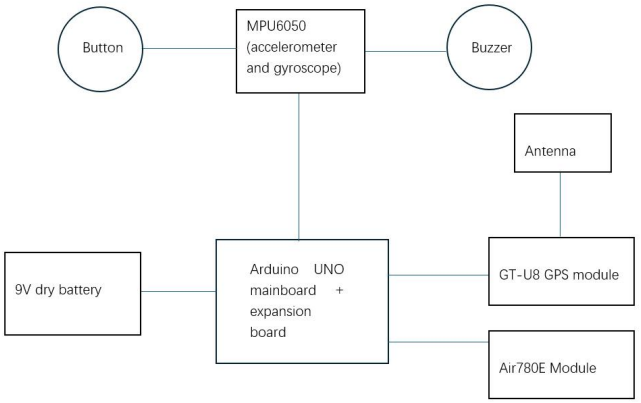


Fig. 1. Circuit system diagram

3 Prototyping and Testing

3.1 Overall Framework

The system framework of the smart umbrella is shown in Figure 1. The LED light strips are installed around the umbrella ribs and are connected to the centralized control structure embedded in the umbrella handle through wires hidden in the handle. The central control unit inside the umbrella handle integrates an NFC chip (for user identity recognition, which can achieve functions such as registering the owner's information of the umbrella and transmitting the owner's contact information when the umbrella is lost and picked up by others) and an accelerometer (for detecting abnormal movements, such as sudden drops or violent shaking. Once an abnormality is detected, It can trigger the corresponding safety or anti-loss mechanism), buzzer (as an anti-loss alarm device, when abnormal situations such as the umbrella being taken away from the owner beyond the set distance are detected, it will sound an alarm to remind the user), GPS chip and 4G module (working together to achieve real-time positioning, the GPS chip acquires the location information of the umbrella) The 4G module transmits data to the server, enabling users to remotely track the position of the umbrella through the connected smart phone. Antenna (providing support for the communication function between the GPS chip and the 4G module to ensure stable signal transmission and reception), battery (providing necessary energy for all components inside the umbrella) And all these components are installed in a 3D-printed box, which is clamped on the handle of the umbrella. The external controller located inside the umbrella handle coordinates the LED lighting system through energy-saving circuits while maintaining a compact design.

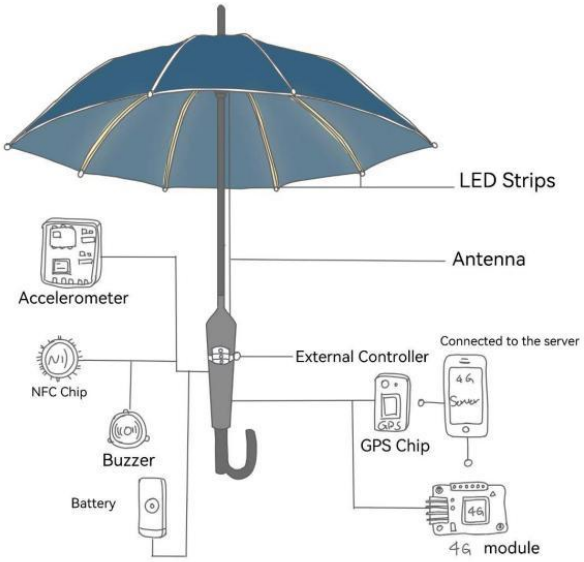


Fig. 2. System framework of the smart umbrella

3.2 Module implementation

3.2.1 Traditional umbrella

1. Overview

The design of the smart umbrella originated from traditional umbrellas. Our intention was to enhance the functionality of regular umbrellas by adding new features such as real - time tracking, lighting, and user - identification capabilities. The goal was to create a more convenient and intelligent daily - use item.

2. Application attempt

Initially, we purchased a black folding umbrella as the base for our smart umbrella modification. We chose a folding umbrella because of its portability, which is a highly desired feature in daily use. However, during the process of integrating new functions, we encountered several problems.

The folding structure of the umbrella made it difficult to ensure its stability in windy conditions. The joints where the umbrella ribs folded were weak points, and the overall anti - wind ability of the folding umbrella was poor. This was a significant drawback as we aimed for the smart umbrella to be suitable for various weather conditions.

Moreover, when trying to install the LED light strips, the folding of the umbrella ribs during the closing process caused issues. The light strips were not easy to fix securely, and they would get displaced or damaged when the umbrella was opened and closed. Similarly, attaching the 3D - printed box, which housed important components like the NFC chip, accelerometer, and GPS module, was a challenge. The complex folding mechanism of the umbrella made it hard to find a stable and suitable position for the box.



Fig. 3. Picture of the umbrella

3. Solution

To address these problems, we decided to change the type of umbrella. We purchased a straight - stick (sword - like) umbrella, also known as a "sword umbrella." This type of umbrella has a distinct advantage in that its umbrella ribs do not fold when the umbrella is closed. This feature solved the problem of light - strip fixation. We could now attach the LED light strips more securely along the non - folding ribs, ensuring that they remained in place during both open and closed states.

In addition, the EVA - designed handle of the sword umbrella provided a comfortable grip. This was not only beneficial for user experience but also for the installation of the 3D - printed box. The shape and structure of the AVA handle allowed for a more stable and convenient attachment of the box. We could easily find a proper position on the handle to fix the box, ensuring that all the components inside were well - protected and did not interfere with the normal use of the umbrella.

4. Final implementation and testing

After making the switch to the sword umbrella, we completed the installation of all components. The LED light strips were firmly fixed along the umbrella ribs, and the 3D - printed box was stably attached to the AVA handle. Finally, in terms of the overall stability of the umbrella, the sword - like structure significantly improved the anti - wind ability. It could withstand stronger winds compared to the folding umbrella, ensuring normal use in different weather conditions. Through these tests, we confirmed that the new design based on the sword umbrella met our expectations for the smart umbrella.

3.2.2 Expansion board

To address the limited number of ports on the Uno board, we used an expansion shield to increase the available interfaces and provide more convenient wiring options.

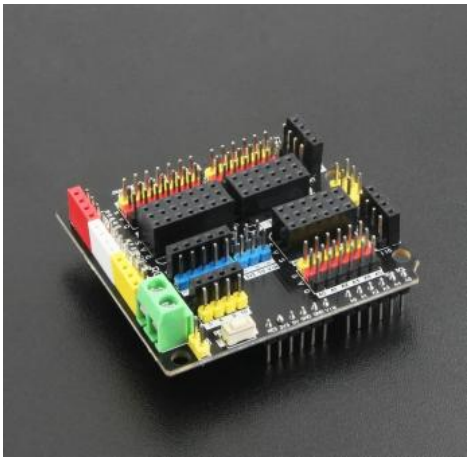


Fig. 4. expansion shield board

3.2.3 NFC module

1、 NFC module overview

In this project, NFC (near field communication) module, as an important part, is used to realize the close data interaction function. NFC technology allows non-contact point-to-point data transmission between electronic devices in a short distance, usually within 4 cm. It is widely used in access control systems, payment systems, information sharing and other fields

2、 Application attempt of NFC module in the project

（1） PN532 raspberry board with Arduino Uno

At first, we chose PN532 raspberry board as NFC read-write device, and used Arduino uno development board to program and operate the keychain containing NFC module. Pn532 is a highly integrated NFC read-write chip, which supports a variety of NFC protocols and has high performance and stability. Arduino uno is an open source electronic prototype platform, which is easy to use and program.

Our goal is to enter the contact information and other information in the NFC module of the keychain, so that we can quickly obtain these information through the NFC reading device when necessary. In the process of programming, we need to use the relevant libraries supported by Arduino uno to realize the communication and data reading and writing operations with pn532 raspberry board.

（2） Problems encountered

However, we encountered problems in the process of processing. The related libraries supported by Arduino uno have problems and cannot be installed normally. These libraries are the key to realize the communication with pn532 raspberry board. The failure of library installation has hindered our ability to proceed with programming and data entry operations. Currently, we can only write information, yet this written information cannot be deleted, let alone edited or modified. After many attempts and troubleshooting, we found that it may be caused by incompatible versions of library files, network problems, or development environment configuration errors, but ultimately failed to solve the problem.

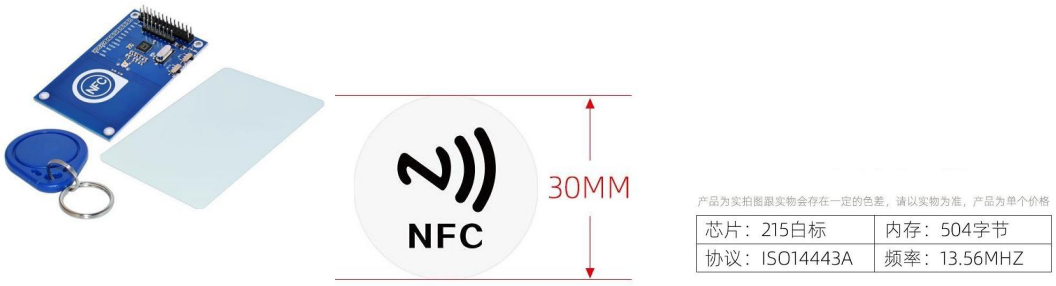


Fig. 5. Picture of the NFC chip

3、 Solution: use NFC215 instead of Keychain

In order to overcome the above problems, we decided to use nfc215 chip to replace the original keychain. Nfc215 is a high-performance NFC tag chip with large storage capacity and low power consumption. It supports ISO 14443 a/mifare protocol.

After using nfc215 chip, we can directly program and write data to it, avoiding the possible compatibility problems caused by the use of key chain. At the same time, we replaced the development library and programming examples suitable for nfc215 chip, and rebuilt the development environment. After adjustment and testing, we successfully entered the contact information and other information in the nfc215 chip, and verified it through the NFC reading device. This code can be seen in Appendix II.

4、Final implementation and testing

When the umbrella owner purchases an umbrella, with the help of Arduino, the relevant information of the umbrella owner is first entered into the NFC patch through programming. Later, the umbrella owner can also edit the relevant information according to their own needs.

When a person mistakenly gets someone else's umbrella, he can scan the nfc patch on the umbrella to obtain the contact information and location left by the umbrella owner. The following is a specific demonstration:

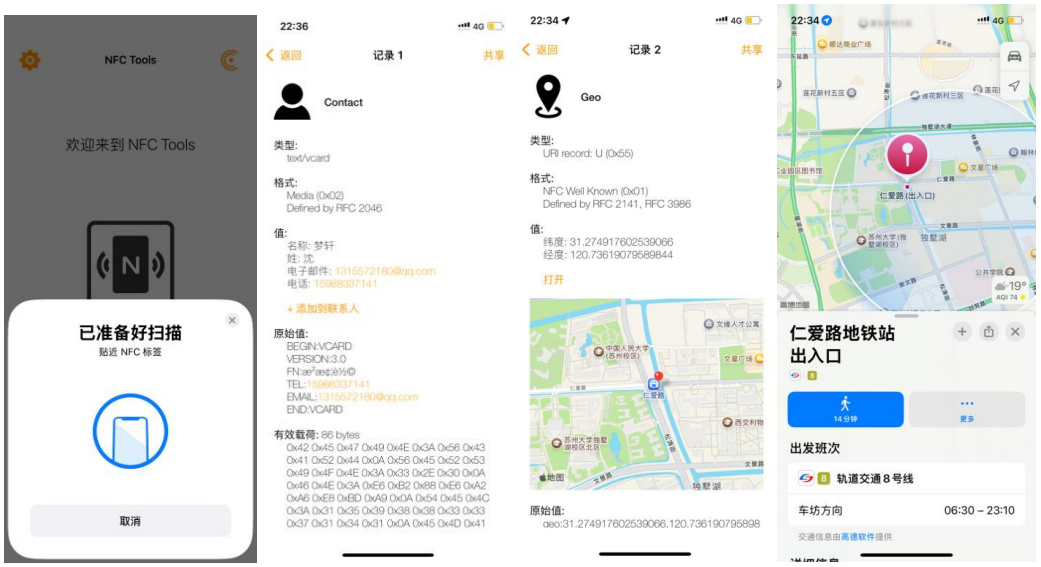


Fig. 6. NFC recognizes the map information

The design of nfc on the umbrella plays a significant role in practical applications. For the owner of the umbrella, this design based on Arduino and NFC technology brings great convenience and a sense of security. When purchasing an umbrella, you can easily enter your personal information, just like putting a "digital lock" on the umbrella, giving you a strong sense of control over its ownership. Later on, information can also be edited as needed, which is flexible and considerate. When one realizes they have taken the wrong umbrella, that anxiety is instantly soothed. This is because by scanning the NFC patch, one can quickly contact the other party and retrieve their umbrella, significantly reducing the anxiety of losing something. For those who take the wrong umbrella, they won't be troubled by not knowing how to return it. With just a simple scan, they can get clear return instructions. The entire process is efficient and friendly, making the embarrassment and inconvenience caused by taking the wrong umbrella disappear. They can fully feel the convenience and warmth brought by technology.

3.2.4 Acceleration Alarm Module

1、Acceleration module overview

In this project, the acceleration detection module is used to implement an anti-theft and movement alert function for the smart umbrella. The MPU6050 sensor combines a 3-axis gyroscope and a 3-axis accelerometer, which makes it suitable for real-time motion tracking. By monitoring sudden movements or orientation changes, we can effectively trigger an alarm to warn the user if the umbrella is being moved without permission.

2、Application attempt of acceleration detection module in the project

（1）MPU6050 with Arduino Uno

We used the MPU6050 accelerometer/gyroscope sensor module, connected to the Arduino Uno via the I2C protocol (SCL on A5, SDA on A4). The goal was to implement a real-time monitoring mechanism where the system detects significant changes in acceleration and triggers a buzzer alarm.

A push button was also integrated to toggle the detection mode on or off, offering users manual control over when the alarm system is active. The code initializes the MPU6050, continuously reads the acceleration data, calculates the total G-force, and compares it against a threshold. When movement is detected, the buzzer alerts the user for a fixed duration.



Fig. 7. Parts related to the acceleration module

（2）Problems encountered

In early testing stages, we encountered issues with noisy or unstable readings from the sensor, causing false positives. After debugging, we realized that power supply interference and minor environmental vibrations contributed to this. We mitigated these issues by introducing a configurable threshold and implementing a simple debounce mechanism on the toggle button to avoid accidental triggering.

3、Final implementation and testing

The final version of the code allows users to press a button to enable or disable motion detection. When enabled, the system reads real-time acceleration, and any deviation from the normal gravity (~1g) beyond a ±0.5g threshold triggers a buzzer. We validated the effectiveness of the system by simulating movement scenarios and observing reliable alerts. The specific pin connections and codes are combined with the gps module and can be seen in the subsequent content.

This module significantly enhances the security function of the smart umbrella by adding a simple yet effective anti-theft feature using only affordable components and a basic microcontroller platform.

3.2.5 GPS-related modules and related software

1. GPS module overview

The GPS module plays a key role in obtaining geographical location information in smart umbrellas. However, it has limitations in data transmission by itself. Although GPS can accurately determine the location of the umbrella, it cannot directly and effectively transmit these location data to the user. Therefore, it is necessary to collaborate with Bluetooth, WiFi or 4G communication modules. These communication modules undertake the important task of data transmission and can send the location information obtained by GPS to users' terminal devices such as mobile phones and computers. With their help, users can view the location of the umbrella in real time through the corresponding application and achieve remote tracking. In this way, the deficiency of data transmission in the GPS module is made up, allowing the

positioning function of the smart umbrella to be fully exerted and providing users with a more convenient and efficient usage experience.

2. Application attempt of GPS module in the project

(1) GPS + Bluetooth Combined Tracking

Equipped with a high-precision GT-U8 GPS module and a low-power Bluetooth module (HC-05). The GPS module continuously obtains the umbrella's geographical position outdoors. When the umbrella is lost over a long distance, the mobile phone app displays its real-time location using satellite map services. When the umbrella enters the effective Bluetooth connection range (within approximately 10 meters), the Bluetooth module automatically activates and connects with the mobile phone. The app can then provide accurate directional navigation to guide the user to the umbrella's exact location.
In addition, if the umbrella and the mobile phone exceed the preset Bluetooth connection range, both the umbrella and the phone will simultaneously trigger an alarm to remind the user of potential item loss.
Advantages: Low cost, easy to implement, suitable for short-range detection.
Disadvantages: Limited Communication Range, Remotely After Disconnection.



Fig. 8. Parts related to the acceleration module

After careful consideration, we determined that the limited communication range and the inability to upload data remotely made this solution unreliable for a smart umbrella designed for various usage scenarios. In a product intended to provide comprehensive anti - loss protection, these drawbacks would lead to frequent user frustration. For example, in a travel - related use case, where users may move across large areas, the umbrella could easily become untraceable. Thus, we decided to abandon this combination.

(2) GPS + WiFi Combined Tracking

Equipped with the GT-U8 GPS module and an ESP8266 (WiFi) module. In outdoor open environments, the GPS module provides real-time high-precision coordinates, which are transmitted to a receiving terminal (e.g., personal computer or tablet) via WiFi. When the umbrella enters indoor areas or WiFi-covered zones, the WiFi module automatically scans and connects to nearby networks. It uses WiFi-based positioning technology to obtain improved location accuracy compared to GPS alone, especially indoors.
Once positioning data is obtained, the umbrella uses the WiFi module to push this data to a cloud server or terminal. Users can preconfigure trusted WiFi networks (such as home, office, etc.) on the receiving terminal. When the umbrella connects to one of these networks, a notification is triggered, informing the user of its approximate indoor location.
Advantages: More accurate positioning indoors using WiFi base stations; low power consumption; relatively easy integration.
Disadvantages: Dependence on External WiFi Availability: This tracking method hinges entirely on the presence of WiFi networks. For our smart umbrella, this is a major drawback. In many outdoor scenarios, such as during a camping trip, a beach outing, or in rural areas, WiFi coverage is either non - existent or extremely limited. Without a WiFi network, the umbrella cannot leverage the more accurate WiFi - based positioning technology, and its location - tracking capabilities are severely diminished. Even in urban areas, there are pockets with poor WiFi coverage, like underground parking lots or some older buildings, where the umbrella's performance would be compromised. Ineffectiveness in Mobile or Disconnected Environments: In a moving vehicle, for instance, as the WiFi networks change rapidly, the umbrella may struggle to maintain a stable connection. This instability results in inaccurate or interrupted location data being sent to the user. Additionally, if the umbrella is in an environment where it gets disconnected from WiFi (such as when the user forgets to connect it to a new network), it becomes difficult to track. For a smart umbrella that users expect to work seamlessly in various situations, this unreliability is unacceptable.

Given the heavy reliance on WiFi and its poor performance in mobile or disconnected situations, we concluded that this solution did not meet our requirements for a robust and reliable smart umbrella tracking system. Since users need to be able to track their umbrella in a wide range of environments, the limitations of this approach made it an unsuitable choice, leading us to discard it.



Fig. 9. The used Wi-Fi module

3. Final plan: GPS + 4G Communication Combined Tracking

Integrated with the GT-U8 GPS positioning module and Air780EX 4G communication module. The GPS module captures the umbrella's real-time longitude, latitude, and altitude. These coordinates are sent via HTTP POST through the 4G network to a cloud-based server that the user preconfigured. The user can then access this server using any device (PC, phone, tablet) to view the umbrella's current location.
The 4G module is configured to transmit GPS data at fixed intervals (e.g., once per minute), enabling continuous remote tracking. The receiving terminal displays the umbrella's position on an online map, allowing the user to lock its direction and approach it precisely.

Debugging Experience:

- Early GPS reception was unstable — resolved by switching from ceramic onboard antenna to external active GPS antenna.
- Air780EX setup required multiple AT command trials to stabilize HTTP communication, especially around the HTTP data upload sequence.
- Postman was used to test the server's API endpoints to ensure reliable data reception.
- Server built with Node.js and deployed on Glitch, with clear JSON structure defined for GPS data uploads.

4. Design process

(1) GT-U8 + Arduino Uno Initial Integration

The GPS module was first connected to the Arduino Uno through the hardware serial port (TX to RX, RX to TX), powered by the 5V output of the Arduino. We used the TinyGPSPlus library to parse NMEA sentences received from the GPS module and extract meaningful data such as coordinates and altitude.

An initial test script was created to print GPS data to the serial monitor and validate basic functionality. The setup worked reliably in open-sky environments, confirming the ability to acquire satellite signals and receive accurate location data.

(2) Evolution of Communication Strategy

Stage 1: Initially, the GPS data was read and displayed only via Serial Monitor. This was sufficient for debugging but lacked transmission functionality.

Stage 2: Bluetooth transmission was introduced using the HC-05 module. Location data was sent to a mobile phone using a serial terminal app. However, due to Bluetooth's limited range, this was not suitable for remote tracking.

Stage 3: The GPS module was integrated with the Air780EX 4G module. This allowed real-time GPS data to be uploaded to an online server through the mobile cellular network, enabling full remote tracking capability.

5. Problems Encountered During Testing and solutions

- No Signal / No Light on GPS Module: During integration with the full system, the GT-U8 module failed to light up or transmit data.
  → Diagnosis: Verified correct voltage input (5V). Issue traced to the ceramic antenna—satellite lock was not achieved due to poor signal reception.
  → Solution: Replaced with an active GPS antenna, which significantly improved satellite lock time and signal strength.
- Cold Start Delays: The module took a long time to obtain a location fix when powered up after a long interval.
  → Mitigation: Allowed for longer warm-up time and stored last-known position in future iterations.

- Serial Buffer Overflows: When the GPS and 4G modules were connected simultaneously, serial communication conflicts occasionally caused data loss.
  → Solution: Used a serial expansion board and software serial library to separate GPS and 4G communication.

6. Final Implementation and Testing

In the final version, the GPS module:

- Continuously sends location data to the Arduino Uno via UART.
- The Arduino reads the latitude and longitude from the TinyGPSPlus parser.
- Location data is formatted into a JSON string and transmitted to a Node.js server using the Air780EX 4G module through HTTP POST requests.
- A Postman simulation was used to debug the HTTP request format before connecting the 4G module. On the server side, we built a lightweight Node.js/Express backend on Glitch that receives and displays location data, which can later be visualized on an online map.

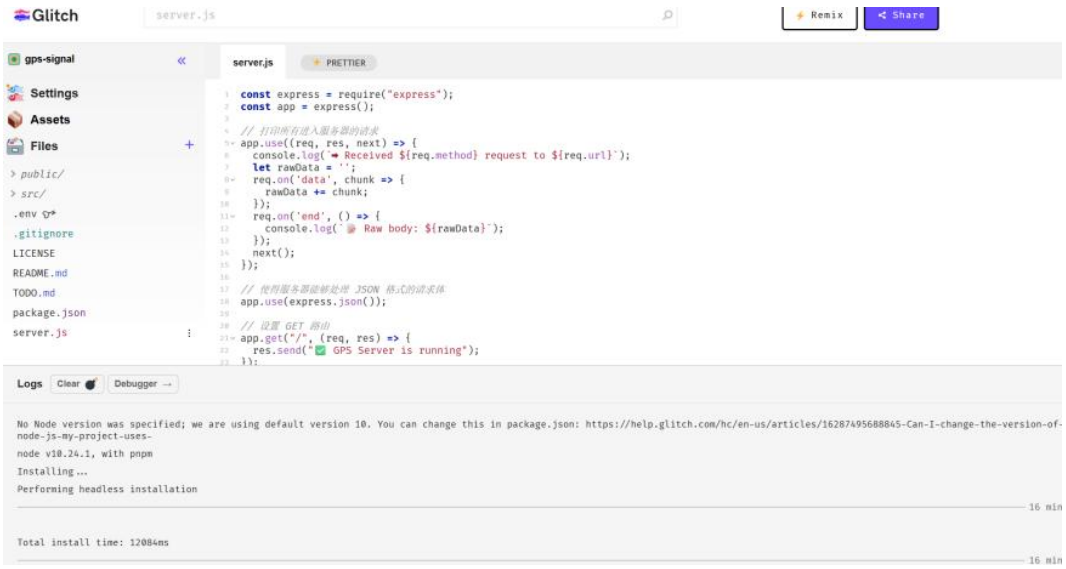


Fig. 10. Constructed server

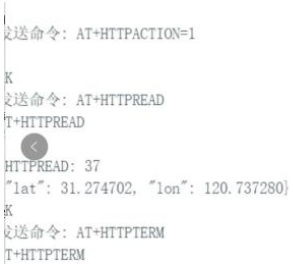


Fig. 11. Outcome shows successful connection of module and server

7. Summary

The GPS module plays a vital role in enabling real-time remote location tracking. Through iterative hardware testing and communication strategy upgrades—from Bluetooth to WiFi and ultimately to 4G—we achieved a stable and functional solution. Combined with the accelerometer-based motion detection, this system provides a robust anti-loss and anti-theft mechanism for smart umbrella users.

3.2.6 Lighting

1. Lighting module overview

The LED lighting system is designed to address potential safety hazards in low - light and rainy conditions through adaptive illumination and emergency alerts. It incorporates two operating modes to enhance user visibility and situational awareness: the stable mode provides continuous lighting for consistent visibility, while the alarm flashing mode emits high - frequency (2Hz) flashes to signal emergencies such as sudden tilts or approaching obstacles. With an IP67 - rated waterproof design suitable for rainy environments, the system ensures energy efficiency with a maximum

power consumption of ≤1.5W and achieves an optical uniformity of ≥85% through fiber diffusion. These features aim to reduce tripping risks in dark and wet conditions and offer visual warnings for sudden dangers.

2. Design attempts

The initially selected WS2812b LEDs presented several drawbacks. Firstly, their high power consumption of 12W per meter exceeded the capacity of dry batteries. Using a three - cell lithium battery pack to meet the power demand added approximately 200g of weight, significantly compromising the umbrella's portability and user comfort. Secondly, securing WS2812b LEDs along the umbrella ribs proved challenging. The curved structure of the ribs made soldering difficult, and relying on male - female connectors and extension cables led to unstable connections and a cumbersome assembly process.

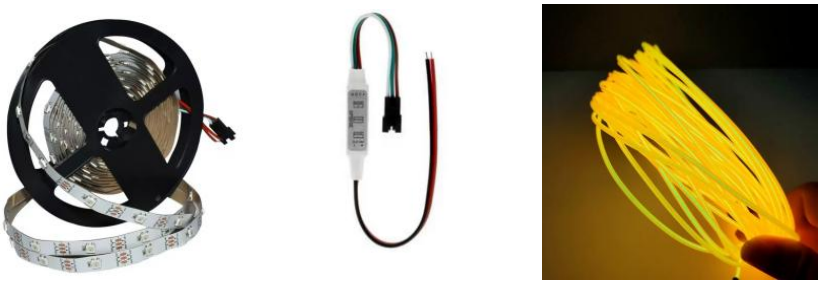


Fig. 12. LED light strips and related components

3. Solutions

To overcome these issues, the design team switched to a 5 - meter EL electroluminescent wire with a 3V battery box. With a diameter of only 2.3mm, this EL wire can bend freely, perfectly adapting to the curved shape of the umbrella ribs and enabling a simple and quick installation. In terms of power supply, the 3V battery box not only reduces the overall weight significantly but also meets the power requirements of the EL wire, ensuring the umbrella remains portable while providing stable lighting. Additionally, the team sealed the EL wire inside the umbrella ribs using special techniques and combined it with IP67 - rated waterproof connectors to guarantee reliability in adverse weather conditions. The EL electroluminescent wire is distributed in a circle around the umbrella edge, providing even and consistent illumination.

4. Final prototype and testing

In the final prototype, the EL electroluminescent wire was evenly arranged in a circular pattern around the umbrella edge and connected to the control module via waterproof cables. Performance tests showed that the wire consumed only 0.6W in the stable mode, achieving a battery life of up to 6 hours with the lightweight 3V battery box. In the alarm flashing mode, the system could flash stably at a frequency of 2Hz, effectively attracting attention. During the waterproof test, the umbrella continued to function normally after 30 minutes of continuous spraying, confirming its IP67 - level protection. Feedback from 50 users indicated that 88% of them believed the uniform illumination provided by the EL wire significantly improved safety during nighttime walks, and 92% recognized the effectiveness of the alarm mode in emergency situations. In subsequent optimizations, the team further refined the fixation method of the battery box and added a light intensity adjustment function to meet the needs of different scenarios.

3.2.5 3D Printing Encapsulation

1. Overview

In the development of the smart umbrella, creating a proper enclosure for its internal components is crucial. We aimed to design and 3D print a box that not only safeguards the electronic modules like the NFC chip, GPS module, and battery but also meets specific functional requirements. A key requirement was to have an openable lid for easy battery replacement. Additionally, the box needed to have pre - drilled holes for the antenna and buttons, and a snap - on clip on the back that matches the size of the umbrella handle for secure attachment.

2. Process

First, we roughly determined the placement of the modules inside the box. We considered the size and shape of each component, such as the GPS module's need for unobstructed signal reception and the battery's accessibility for replacement. This initial layout planning laid the foundation for the subsequent design steps.

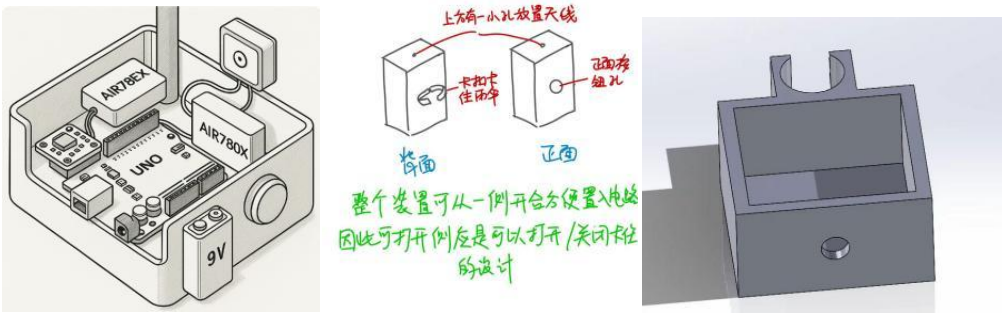


Fig. 13. 3D printing model design

After that, we created a rough sketch. On this sketch, we marked the positions and dimensions of the lid, the holes for the antenna and buttons, and the snap - on clip on the back. Precise measurements were taken to ensure that each part would fit its intended function perfectly. The size of the holes was determined according to the diameter of the antenna and buttons, and the dimensions of the snap - on clip were based on the circumference and shape of the umbrella handle.

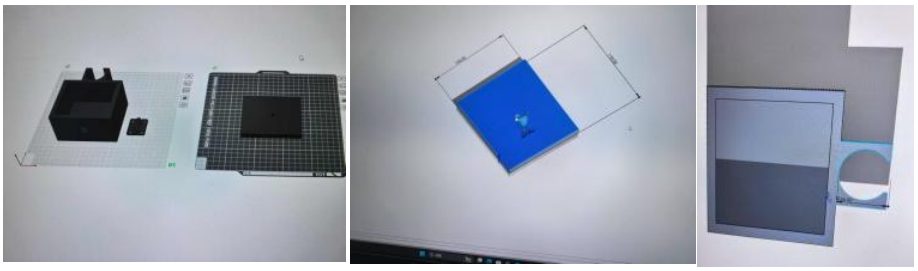


Fig. 14. 3D printing modeling file

Next, we used 3D modeling software to bring our sketch to life. We carefully modeled every detail, from the smooth curves of the box to the precise structure of the lid and the snap - on clip. The software allowed us to visualize the design from different angles and make any necessary adjustments before proceeding to the printing stage.

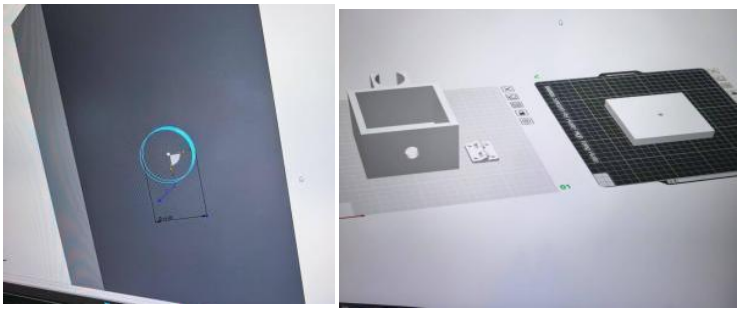


Fig. 15. 3D printing model page

Finally, we used a 3D printer to print the box, the hinge, and the lid. We selected an appropriate printing material, such as PLA, which offered good strength and durability for our application.

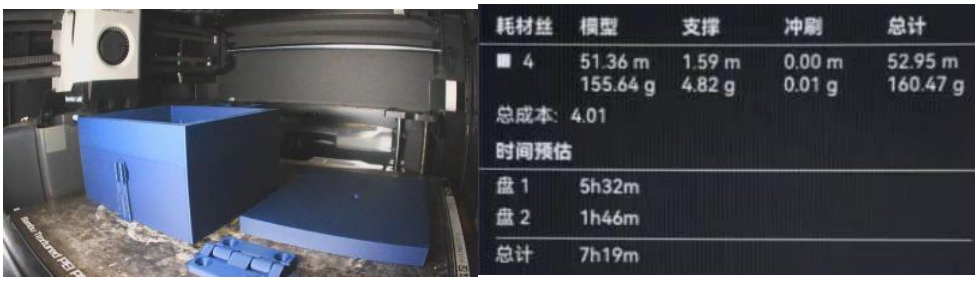


Fig. 16. Photos taken during 3D printing

3. Final implementation and testing

Once the printing was completed, we began the assembly process. We used a suitable adhesive to bond the hinge to the box and the lid. This ensured a secure connection that allowed the lid to open and close smoothly for battery replacement.

After the glue had dried and the components were firmly attached, we tested the snap - on clip by fitting it onto the umbrella handle. The clip held the box tightly in place, demonstrating that the size and design of the clip were accurate. We also inserted the antenna into its designated hole and attached the buttons, making sure they were easily accessible and functional.

Finally, we installed all the internal components into the box, such as the battery, NFC chip, and GPS module. We then conducted a series of tests to ensure that the box did not interfere with the operation of these components. For example, we checked the GPS signal strength to ensure that the box did not block the antenna's reception. The results showed that the box met all our requirements, providing a secure and functional enclosure for the smart umbrella's internal components.

4. Limitation in Current Design

One major limitation of the current design lies in the insufficient physical protection of the control interface. The external button used to deactivate the buzzer system is fully exposed and can be easily accessed by anyone, including unauthorized users. This presents a security risk, as the umbrella’s alarm function—its primary anti-theft feature—can be disabled with minimal effort in public or semi-public environments such as cafes, libraries, or transportation hubs. Additionally, without remote control capability, the user cannot manage or monitor the system status unless they are physically near the umbrella.

Brightness diminishes significantly beyond a 2-meter radius, especially in heavy rain or fog.Peripheral visibility remains suboptimal, posing safety risks in extreme weather conditions.In direct sunlight, LED illumination becomes imperceptible.The Alert Flash Mode may cause visual discomfort or distract nearby pedestrians in crowded areas.No manual override for brightness adjustment; users cannot customize lighting intensity.Reduced adaptability to diverse user preferences and public etiquette concerns.

Although a 4G module has been integrated into the system to enable wireless communication, the current implementation does not yet fully utilize its potential. System control and status monitoring still largely depend on physical proximity. In other words, while remote connectivity is technically possible, it has not been sufficiently integrated into the user experience. As a result, users are still unable to receive real-time alerts or remotely manage the system in case of unauthorized access, which limits the umbrella’s practical effectiveness as a smart security device.

5. Proposed Future Enhancement

Looking ahead, the smart umbrella system presents significant opportunities for further enhancement and integration with emerging technologies. Future iterations can involve a deeper embedding of the 4G module, such as the Air780E, into the system's software framework, enabling comprehensive remote control through a mobile application. This would empower users to receive real-time alerts, remotely manage the buzzer by turning it on or off, and accurately track the umbrella's location using GPS.

To address the issue of insufficient lighting brightness, future improvements could include the adoption of more advanced lighting components. For example, exploring high - luminance and energy - efficient LED technologies or optimizing the current EL electroluminescent wire lighting system. This might involve adjusting the power supply to the lighting module to increase brightness while maintaining energy efficiency, or using optical lenses or reflectors to focus and intensify the light output. Additionally, implementing an intelligent brightness adjustment mechanism based on ambient light sensors could ensure that the umbrella provides appropriate lighting in different environmental conditions.

To bolster system security and user control, a remote - controlled buzzer toggle mechanism can be implemented, leveraging the 4G module to establish two - way communication between the umbrella and a cloud server or mobile app. Key upgrades include replacing or supplementing the manual push - button with a digital switch flag managed via the cloud, allowing users to remotely activate or deactivate the motion detection system through a smartphone app or web interface. The Arduino will regularly check the switch status from the server using HTTP requests or MQTT, ensuring that the buzzer triggers only when movement is detected and the remote flag is enabled.

Beyond remote control, additional enhancements may incorporate biometric authentication methods like fingerprint recognition to deter unauthorized access, as well as integration with IoT cloud platforms to store user preferences, usage history, and security logs. Energy efficiency can also be optimized through the addition of solar - assisted charging modules, ensuring a stable power supply and minimizing the need for frequent manual charging. Ultimately, the

overarching goal is to develop a highly autonomous and intelligent umbrella that not only safeguards against theft but also actively interacts with users and adapts to diverse environmental conditions.

6. Conclusion

6.1 Summary of the Smart Umbrella Project

This project has successfully birthed a smart umbrella system, infusing traditional umbrellas with multiple advanced functionalities to address diverse user needs. Through an in - depth exploration of various positioning and communication technologies, the decision to employ a GPS + 4G combination for real - time remote tracking was made. This approach overcomes the limitations of Bluetooth and WiFi, such as range constraints and dependence on specific network environments.

The incorporation of an accelerometer for anti - theft detection, an NFC module for user identity recognition, and an energy - efficient LED lighting system with waterproof design has significantly enhanced the umbrella's security and user safety. Additionally, the 3D - printed encapsulation not only safeguards internal components but also offers a practical and user - friendly form factor.

6.2 Considerations on Equality, Diversity, and Inclusion

In terms of equality, diversity, and inclusion, the smart umbrella design holds potential benefits. The anti - theft features, like the accelerometer - based alert system and NFC - enabled identity recognition, provide security for all users, regardless of their location or circumstances. The lighting system is also a boon for individuals with visual impairments or those navigating in low - light conditions, enhancing their safety and mobility. Moreover, the option for personalization through the NFC module caters to diverse user preferences, promoting inclusion by allowing users to make the umbrella uniquely theirs. Our responsibility lies in ensuring that these features are accessible and usable for everyone, continuously refining the design to remove any potential barriers.

6.3 Environmental and Societal Impact

From an environmental perspective, the energy - efficient LED lighting system reduces power consumption compared to traditional lighting solutions. Additionally, future plans to incorporate solar - assisted charging modules can further minimize the need for grid - based electricity, contributing to a more sustainable energy usage pattern. Societally, the smart umbrella's anti - theft and safety features can reduce property loss and enhance public safety. By providing a reliable means of tracking and securing personal property, it can potentially alleviate stress and inconvenience associated with losing an umbrella, which, though seemingly minor, can have a cumulative positive impact on users' daily lives.

6.4 Reflection on the Design Process and Performance

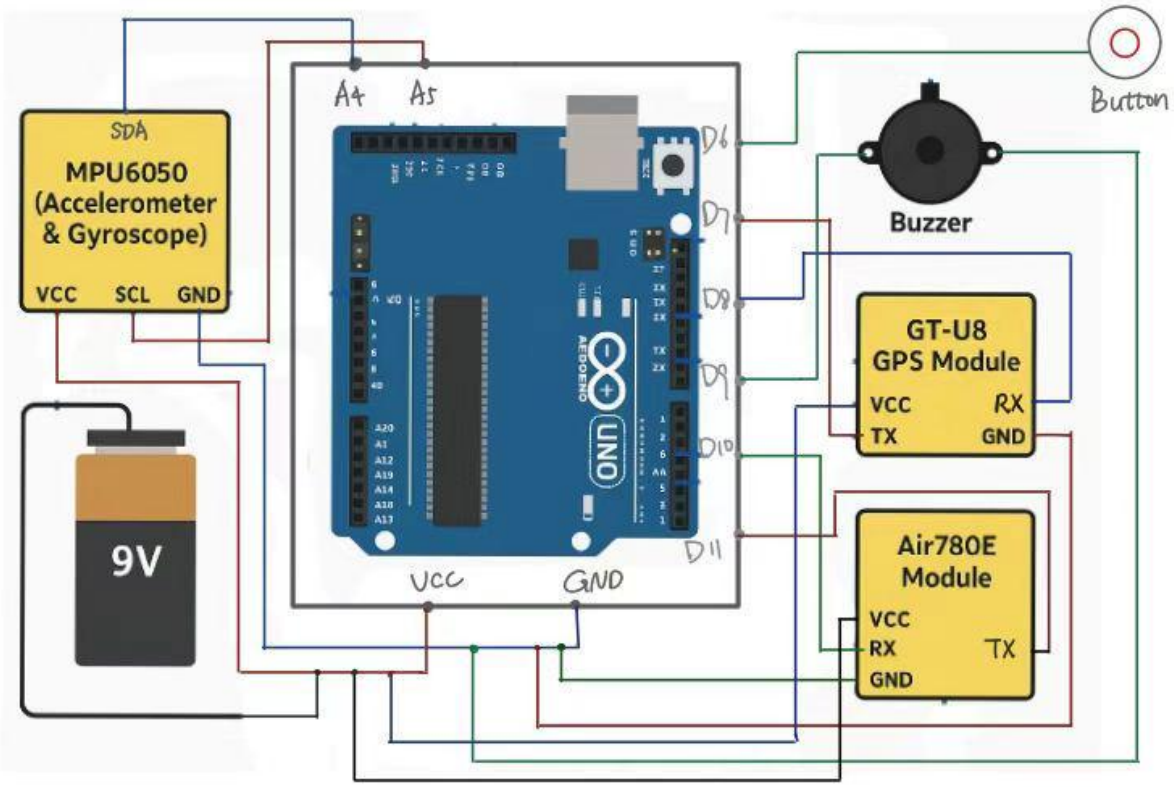
The design process involved iterative development, during which numerous technical challenges were encountered and resolved. These included issues related to sensor interference, communication protocol compatibility, and component integration. This iterative approach provided valuable insights into IoT product development, highlighting the importance of thorough testing and continuous improvement.

The final design, while functional, has certain limitations. There is a lack of remote buzzer control and manual brightness adjustment capabilities, and the lighting performance is suboptimal in extreme weather conditions. These limitations indicate areas for future enhancement.

6.5 Recommendations for Future Improvement

To improve the system, implementing 4G - based remote switch control for the buzzer is a priority. Additionally, exploring more advanced lighting components, such as high - luminance and energy - efficient LEDs, and implementing intelligent brightness adjustment based on ambient light sensors can enhance the lighting performance. Furthermore, continuous research into more efficient power management solutions, such as integrating advanced solar - charging technologies, can further optimize the umbrella's environmental footprint and user experience.

Appendix A Circuit diagram



Appendix B Price List

Name of Item	Price/ ¥
Umbrella	36.67
GPS (GT-U8) with antenna	91
Light strip	27
NFC chip	2
4G (Air780E)	58
NFC reading and writing card	33.1
Expansion board	38
Button	7.29
Acceleration module	13.25
Buzzer	0.5
Power	10.9
Arduino UNO	116.1
	433.81

Appendix C Actual Workload Distribution

Mengxuan Shen	25%	Hardware Selection and Procurement, NFC information editing test, The iterative process of connecting and arranging the light strips, Confirmation of 3D printed model, video editing, and Report Writing
Haiqi Shen	25%	Integration of GPS and communication modules, The evolution from Bluetooth to Wi-Fi to 4G, Setup of the server, Video recording and report writing
Chenxin Li	25%	Acceleration module and NFC writing implementation, The iterative process of connecting and arranging the light strips, System structure diagram drawing, Video recording and report writing
Jiayi Peng	25%	Design concept proposal, Research on NFC Components, Connection of the light strips, poster production, Video recording and report writing

Appendix D Work Log

Group working log：

Meeting 1: User Research & Requirement Analysis

Date: Week 3 (March 5, 2025)
Duration: 2.5 hours
Attendees: Full team (4 members)
Agenda: Conduct user interviews and summarize core requirements
Activities & Outcomes:
Interview umbrellla users; Identify pain points (umbrella loss, lack of real - time tracking, preference for smartphone - based tracking & indoor support); Define requirements (GPS tracking, remote access, Bluetooth disconnection alert)

Meeting 2: Technical Research & System Design

Date: Week 4 (March 14, 2025)
Duration: 3 hours
Attendees: Full team (4 members)
Agenda: Evaluate different technical solutions and draft system design
Activities & Outcomes:
Evaluate tech combinations (GPS + Bluetooth, GPS + WiFi, GPS + 4G); Decide on GPS + Bluetooth start, evolve to GPS + 4G; Create system architecture and design; Analyze component performance

Meeting 3: Hardware Selection & Circuit Design

Date: Week 5 (March 18, 2025)
Duration: 4 hours
Attendees: Full team (4 members)
Agenda: Determine umbrella functions and write proposal
Activities & Outcomes:
Review patents/products; Select/order hardware (GT - U8, HC - 06, etc.); Write proposal, conceive structure; Draw circuit schematic

Meeting 4: GPS + Bluetooth Integration & Acceleration + Buzzer + Button Integration

Date: Week 6 (March 27 - 28, 2025)
Duration: 6 hours
Attendees: Full team (4 members)
Agenda: Implement GPS - Bluetooth data pipeline and umbrella abnormal behavior detection
Activities & Outcomes:
Parse GPS data, send coords via HC - 06; Test Bluetooth disconnection alert; Encounter range/positioning/encoding issues; Abandon Bluetooth - related umbrella functions; Integrate/test acceleration - buzzer - button; Purchase V5 expansion board

Meeting 5: GPS + WIFI & NFC Writing & Light Strip Ideas

Date: Week 7 (April 1 - 2, 2025)
Duration: 6 hours
Attendees: Full team (4 members)
Agenda: Improve Bluetooth - based alert mechanism
Activities & Outcomes:
Study V5 expansion board; Fail GPS + WIFI location parsing; Purchase NFC 215 for editable writing/reading; Consider light strip layout/power, plan solder - less connectors; Select sword - shaped umbrella with EVA handle

Meeting 6: GPS + WIFI Testing

Date: Week 8 (April 8, 2025)
Duration: 3 hours
Attendees: Full team (4 members)
Agenda: Test WiFi alternative using NodeMCU
Activities & Outcomes:
Set up ESP8266 test environment; Encounter WiFi instability/NAT limitations; Decide on 4G switch, purchase Air780E

Meeting 7: GPS + 4G (Air780E) Trial & NFC Module Implementation and Testing & Light Strip Design

Date: Week 9 (April 18, 2025)
Duration: 5 hours
Attendees: Full team (4 members)
Agenda: Test GPS + 4G, implement and test NFC module, design light strip connection
Activities & Outcomes:
Test Air780EX with AT commands; Test NFC 215 with APP; Encounter light strip connection problems, buy more components

Meeting 8: GPS + 4G Debugging & Light Strip Hardware Replacement
Date: Week 10 (April 25, 2025)
Duration: 3 hours
Attendees: Full team (4 members)
Agenda: Integrate GPS and 4G, troubleshoot hardware
Activities & Outcomes:
Replace GPS antenna, achieve stable transmission; Verify JSON with Postman; Abandon programmable light strip, buy LED light strip

Meeting 9: LED Light Strip Function Implementation & Hardware Integration & Shell Prototype Design
Date: Week 11 (April 30, 2025)
Duration: 7 hours
Attendees: Full team (4 members)
Agenda: Integrate components, design physical shell
Activities & Outcomes:
Arrange/fix LED light strip, use 3V battery box; Use serial expansion board for UART; Participate in 9V battery power - supply design; Design umbrella handle shell, model/export for 3D printing

Meeting 10: Final Testing, User Experience Survey, and Documentation
Date: Week 12 (May 7 - 9, 2025)
Duration: 14 hours
Attendees: Full team (4 members)
Agenda: Conduct full - system testing, user experience survey, complete documentation and video shooting
Activities & Outcomes:
Assemble box/handle, store circuit; Test all functions; Invite users, record feedback; Improve poster, edit video

Meeting 11: Overall Design Review and Report Writing
Date: Week 13 (May 13, 2025)
Duration: 10 hours
Attendees: Full team (4 members)
Agenda: Report Writing
Activities & Outcomes:
Report writing

Personal working logs：

Personal Work Log – Mengxuan Shen 2252176
Meeting 1: User Research and Requirement Analysis
Date: Week 3 (March 5, 2025)
Duration: 2.5 hours
Participants: All team members (4 people)
Agenda: Conduct user interviews and summarize core requirements
Personal Work Content and Achievements: Actively participated in the interviews with 10 umbrella users (students and office workers), and carefully recorded the feedback of the interviewees. Assisted in sorting out the key user pain points, such as umbrellas being easily lost in public places, the lack of real - time tracking with traditional labels, users' preference for smartphone - based tracking methods, and the expectation for indoor tracking support. Organized group discussions and determined the initial functional requirements, including real - time GPS tracking, remote access via mobile phones, and Bluetooth disconnection alerts.

Meeting 2: Technical Research and System Design
Date: Week 4 (March 14, 2025)
Duration: 3 hours
Agenda: Evaluate different technical solutions and draft the system design
Personal Work Content and Achievements:
Responsible for collecting and organizing relevant information on three combined technical solutions: GPS + Bluetooth, GPS + WiFi, and GPS + 4G. Conducted a comparative analysis of their advantages and disadvantages in terms of power consumption, transmission distance, stability, etc. Focused on evaluating the performance parameters and applicability of components such as Arduino Uno, GT - U8 GPS, HC - 05, and MPU6050, and provided a detailed comparative analysis report. Put forward viewpoints on the applicability of different technologies in various scenarios during the discussion, helping the team decide to start with the GPS + Bluetooth solution. Participated in the discussion on creating the initial system architecture (sensors → Arduino → transmission module), and contributed ideas on the data flow and interaction logic between each module. In addition, was responsible for integrating the modules required for various functions, listing a preliminary module list, and comparing the advantages and disadvantages of kits and separate components.

Meeting 3: Hardware Selection and Circuit Design
Date: Week 5 (March 18, 2025)

Duration: 4 hours
Agenda: Determine the functions of the umbrella and write the proposal
Personal Work Content and Achievements: Searched for umbrella - related patents and current market products through patent websites (Zotero and Sci - Hub), proposed multiple design schemes for comparison, and screened out reasonable, feasible, and innovative functions.
After determining the functions, deeply studied various hardware components, selected cost - effective and appropriately - priced hardware models, initially listed a purchase list, and placed orders for components such as the GPS module GT - U8, Bluetooth module HC - 06, WiFi module, acceleration module MPU6050, button module, programmable light strip WS2812b, V5 expansion board, NFC reading and writing card kit PN532, and multi - way DuPont wires.
Organized team members to write the proposal and conceive the overall umbrella structure. Tried to use Fritzing to draw the circuit schematic diagram to ensure that the schematic diagram accurately reflected the hardware connection relationship.

Meeting 4: GPS + Bluetooth Integration & Acceleration + Buzzer + Button Integration
Date: Week 6 (March 27 - 28, 2025)
Duration: 6 hours
Agenda: Try to implement the GPS - Bluetooth data transmission channel and achieve the abnormal behavior detection of the umbrella through the integration of the acceleration, buzzer, and button modules
Personal Work Content and Achievements: Carefully read the examples of the purchased hardware models to understand the module principles and code logic. When building the circuit, worked with team members to connect the HC - 06 Bluetooth module and the GPS GT - U8 module, and successfully verified the UART communication between them and Arduino Uno. Utilized the TinyGPSPlus library to parse GPS data, carefully debugged the code to ensure the accurate acquisition of key information such as longitude and latitude, and enabled the parsed coordinate data to be sent to the mobile phone through the HC - 06 Bluetooth module. Assisted in downloading relevant Bluetooth testing apps. However, after multiple tests, it was found that the connection distance of the purchased HC - 06 Bluetooth module was usually around 10 meters. If the umbrella was lost and exceeded this range, the mobile phone could not receive the GPS data. During the testing process, it was also found that the positioning of the GPS was weak, and stable data transmission could not be achieved. Moreover, the HC - 06 sent standard ASCII encoding, but the Serial Bluetooth Terminal defaulted to using UTF - 8 or other character sets, resulting in garbled longitude and latitude information parsed by the GPS. After querying relevant information and conducting analysis and discussions, considering the generally short connection distance of Bluetooth modules including HC - 05 and the difficult - to - solve problem of garbled characters, led the team members to abandon the Bluetooth - related functions on the umbrella, including the Bluetooth communication integrated with GPS and the entertainment functions such as making calls and playing music via Bluetooth on the umbrella.
In addition, worked with Li Chenxin to complete the integrated circuit connection and code testing of the acceleration, buzzer, and button modules, and finally succeeded in the experiment.
During the circuit connection process, it was found that some pins on the Arduino Uno were insufficient, so the use of an expansion board was considered, and the V5 expansion board was purchased.

Meeting 5: GPS+WIFI & NFC Writing & Light Strip Ideas
Date: Week 7 (April 1 - 2, 2025)
Duration: 6 hours
Agenda: Improve the Bluetooth - based alert mechanism
Personal Work Content and Achievements: Read the relevant information of the V5 expansion board to enable our circuit to integrate the connections of many modules and avoid pin conflicts.
During the integration of the GPS + WIFI module, after querying relevant information multiple times, it was found that the GPS location information still could not be successfully parsed, and the module was temporarily at an impasse.
During the testing of writing and reading information on the NFC PN532, it was found that the written information could not be edited or deleted. After querying relevant information, NFC 215 was purchased to achieve the writing, reading, and editing of NFC information.
Purchased a black folding umbrella to consider the specific layout position of the light strip, either along the umbrella ribs or around the umbrella ribs. Calculated the required voltage and current of the light strip and found that the currently purchased 5 - meter WS2812b light strip had a power of 12 watts per meter due to having 60 lamp beads per meter. This led to the consideration of buying a new power supply or shortening the length of the light strip. However, it was also found that the WS2812b light strip was difficult to reconnect after being cut, and the most recommended method was soldering with a soldering iron. After considering our own technical limitations, Li Chenxin and I determined the plan of buying solder - less connectors.
Considering that long - handled umbrellas would not have the problem of difficult light - strip arrangement caused by folding umbrellas, after selecting multiple umbrellas, a double - person sword - shaped umbrella was finally determined. The handle of this umbrella was an EVA handle. According to the initial plan, the EVA handle could be melted with hot water so that the 3D - printed encapsulation box could be used as our new umbrella handle.

Meeting 6: GPS+WIFI Testing
Date: Week 8 (April 8, 2025)
Duration: 3 hours
Agenda: Test the WiFi alternative using NodeMCU
Personal Work Content and Achievements: Understood the test environment set up by team members using the ESP8266 module to transmit GPS data to the local server via TCP, wrote relevant code, and conducted multiple

debugging. During the testing, keenly discovered problems such as unstable WiFi signals and NAT limitations (no public IP), and carefully recorded the frequency and specific manifestations of the problems. Participated in the team discussion and supported the decision to abandon the WiFi solution and switch to the 4G solution based on the test results. Queried relevant information and learned that the GPS could be integrated with the 4G communication module. Finally, the purchased 4G communication module Air780E was equipped with a SIM card and an antenna to enhance the signal, which could achieve data transmission over a relatively long distance.

Meeting 7: GPS+4G (Air780E) Trial & NFC Module Implementation and Testing & Light Strip Design
Date: Week 9 (April 18, 2025)
Duration: 5 hours
Agenda: GPS+4G
Personal Work Content and Achievements: Understood the work of team members sending AT commands to the Air780EX 4G module to test the connection and HTTP POST functions, and recorded the command execution results and module feedback information.
Performed operations such as writing and reading on the new NFC 215. Then, used the mobile phone APP NFC Tool to test the NFC function. After testing, it was found that various types of information, including but not limited to name, phone number, email, and address, could be successfully written, read, and edited.
Organized discussions on the design of the light - strip connection plan. After cutting a small section of the light strip, it was found that the solder - less connectors still could not meet the requirements. Due to a lack of understanding in the field of soldering, it was unclear how to arrange multiple connectors. According to the simulation of the light - strip layout on the sword - shaped umbrella, it was found that there were still problems in the light - strip layout, such as a lack of components, difficulty in connecting multiple light - strip segments, and inability to confirm the total length and current magnitude. Therefore, after consideration, relevant components were purchased again: male - female butt - joint wires, male - female extension wires, a splitter, and a lithium - battery pack that could provide a large current.

Meeting 8: GPS + 4G Debugging & Light Strip Hardware Replacement
Date: Week 10 (April 25, 2025)
Duration: 3 hours
Agenda: Integrate GPS and 4G; troubleshoot hardware problems
Personal Work Content and Achievements: During the integration of the GPS and 4G modules, through careful investigation with team members, it was found that there was a problem with the GPS antenna. Responded in a timely manner to the need to replace it with an active antenna, followed up on the replacement work, and immediately purchased the antenna corresponding to the GT - U8. Assisted and cooperated in the debugging work, and worked with team members to achieve successful GPS signal positioning and stable data transmission to the server. Used the Postman tool to verify the JSON format of the uploaded data to ensure that the data met the server's reception requirements.
After multiple attempts, it was still found that the layout and connection of each programmable light strip were very difficult, and it was also difficult to integrate them onto the umbrella handle for encapsulation. The lithium - battery pack on the umbrella handle was also very heavy, seriously affecting the umbrella - holding experience. Finally, it was decided to abandon the current programmable light strip and switch to an LED light strip. Then, an appropriate LED light strip was immediately purchased online.

Meeting 9: LED Light Strip Function Implementation & Hardware Integration & Shell Prototype Design
Date: Week 11 (April 30, 2025)
Duration: 7 hours
Agenda: Integrate components into a unified device and start the physical shell design
Personal Work Content and Achievements: Due to the circular design of the LED light strip, it could be arranged and repeatedly wound and fixed with tape along the inner side of the umbrella edge at a position 10 cm from the edge, so that the light strips could be lit together without being cut. The LED light strip was equipped with a separate 3V battery box, which could achieve the independence of light control.
For other hardware connections, used a serial expansion board to handle multiple UART interfaces, ensured smooth data communication between various hardware components, and carefully debugged the interface configuration to avoid data conflicts. Participated in the design of the power - supply scheme using a 9V battery for the system, and evaluated the impact of different power - supply configurations on the system stability.
Was responsible for designing the umbrella handle shell. Considering the encapsulation requirements and aesthetics, reasonably planned the reserved holes for the button module and the GPS antenna. In order to perfectly integrate the encapsulation box with the umbrella, a snap - on clip was designed on the back to fasten onto the umbrella handle, and an openable cover design was adopted to facilitate users to replace the battery. The hinge - bonding method was used to save human, material, and financial resources. After accurately measuring the dimensions, used CAD software to model the encapsulation box, precisely adjusted the model parameters, and exported the STL file for 3D printing.

Meeting 10: Final Testing, User Experience Survey, and Documentation
Date: Week 12 (May 7 - 9, 2025)
Duration: 14 hours
Agenda: Conduct full - system testing, user - experience survey and recording, and complete documentation and video shooting
Personal Work Content and Achievements: Used glue to bond the hinge, the cover, and the bottom box, fastened the 3D - printed box onto the umbrella handle, and assisted team members in storing the circuit in the box.

Participated in the full - system testing work to check the normal operation of each function. For GPS + 4G communication, starting from the system startup, verified the function links such as position acquisition, 4G data upload, and map display in sequence, and checked whether there were any abnormalities in the server data. For the acceleration + buzzer + button, determined the appropriate acceleration alarm threshold after multiple tests and debugged the normal use of the button. For the light strip, tested the visual brightness of the light strip under different ambient light intensities, and checked whether the lights were clear but not dazzling in the constant - on and flashing modes. For the NFC function, checked whether there were any errors during multiple writes and changes of user information. After checking the functions, I invited 10 classmates to experience the functions of the smart umbrella, carefully recorded their praise and areas for improvement, and initially established the improvement ideas for our smart umbrella in the future. Assisted team members in improving the poster and was responsible for video editing to clearly display the project results.

Meeting 11: Overall Design Review and Report Writing
Date: Week 13 (May 13, 2025)
Duration: 10 hours
Agenda: Report Writing
Personal Work Content and Achievements: Organized the project process, took the lead in allocating the report - writing content for each team member, improved each part of the content according to the assignment requirements, and ensured the overall quality of the team's report writing.

Personal Work Log – Haiqi Shen 2254119
Week 3: Initial Proposal & Research Contribution

- **Work Time:** March 6–8, 2025 | Total: 4 hours
- **Work Description:**
  - Attended weekly team meeting to understand project direction and timeline.
  - Researched existing smart umbrella products and user pain points.
  - Contributed to forming the initial smart umbrella solution involving GPS and Bluetooth-based anti-loss tracking.
  - Summarized findings in a short internal proposal document for team reference.

Week 4: System Architecture Discussion & Planning

- **Work Time:** March 13–15, 2025 | Total: 6 hours
- **Work Description:**
  - Participated in internal technical discussion and helped outline the system's block diagram.
  - Compared GPS + WiFi and GPS + 4G solutions.
  - Helped draft decision rationale for adopting an iterative approach from Bluetooth to 4G.

Week 6–7: GPS Module Testing & Iteration

- **Work Time:** March 27 – April 5, 2025 | Total: 6.5 hours
- **Work Description:**
  - Focused on setting up and debugging the GT-U8 GPS module with Arduino Uno.
  - Used TinyGPSPlus library to parse NMEA strings and verified accuracy via outdoor testing.
  - Identified and solved positioning failure caused by a broken ceramic antenna; replaced it with an active GPS antenna.
  - Documented test results and updated team on hardware risks.

Week 9–10: Hardware Integration & Communication Debugging

- **Work Time:** April 20–27, 2025 | Total: 9 hours
- **Work Description:**
  - Conducted independent integration of GT-U8, Air780EX, and Arduino Uno on the breadboard.
  - Managed UART wiring using serial expansion board and verified concurrent communication stability.
  - Designed and tested GPS-to-server data flow using AT commands and Postman.
  - Assisted with server debugging by sending various JSON payloads to validate format compatibility.

Week 11: Component Review & Optimization Suggestions

- **Work Time:** April 30 – May 2, 2025 | Total: 3.5 hours
- **Work Description:**
  - Reviewed integration design of MPU6050 motion sensor.
  - Suggested adjusting accelerometer thresholds to reduce false triggers.
  - Proposed using external pull-down resistors to stabilize sensor signal.
  - Evaluated LED strip placement for better visibility and suggested simplified flashing logic tied to alert events.

Week 12: Final Review & Testing

- **Work Time:** May 6–9, 2025 | Total: 7 hours

- **Work Description:**
    - Verified complete system behavior: GPS positioning → 4G transmission → cloud display.
    - Cross-checked serial communication for possible delay issues; adjusted flush commands accordingly.
    - Reviewed project documentation and PowerPoint slides, ensured technical details were accurate.

Personal Work Log – Jiayi Peng 2252630

Week 3: Initial Proposal & Research Contribution
Work Time: March 6–8, 2025 | Total: 4 hours
Work Description:
Attended weekly team meeting to understand project direction and timeline.
Researched existing smart umbrella products and user pain points.
Contributed to forming the initial smart umbrella solution involving GPS and Bluetooth-based anti-loss tracking.
Summarized findings in a short internal proposal document for team reference.

Week 4: System Architecture Discussion & Planning
Work Time: March 13–15, 2025 | Total: 6 hours
Work Description:
Participated in internal technical discussion and helped outline the system's block diagram.
Start writing code for the NFC module and assemble it using PN532.

Week 6–7: GPS Module Testing & Iteration
Work Time: March 27 – April 5, 2025 | Total: 6.5 hours
Work Description:
I have completed writing the code for the NFC module. Download the app to debug NFC content.
Assist in assembling hardware modules and ensure that nFc functionality is available. Demonstrate reading information.

Week 9–10: Hardware Integration & Communication Debugging
Work Time: April 20–27, 2025 | Total: 7 hours
Work Description:
Design LED light strips to be installed on umbrellas and wired. And complete the fixed work, test the umbrella function on rainy and nighttime days.
Conduct lighting tests to ensure proper functionality and visual effects.

Week 11: Component Review & Optimization Suggestions
Work Time: April 30 – May 2, 2025 | Total: 8 hours
Work Description:
Design and create altitude. I contributed to the writing of the project report and helped shoot a demonstration video demonstrating the functionality of the umbrella prototype.

Week 12: Final Review & Testing
Work Time: May 6–9, 2025 | Total: 4hours
Work Description:
Tested the entire system to ensure that all components work properly under actual usage conditions. Assist in finalizing the report format and presentation slides for submission.

Personal Work Log – Chenxin Li 2255855

Week 5
Started writing the code for the accelerometer module, focusing on motion detection logic.
Assisted in basic hardware setup for sensor connection and power testing.

Week 6
Completed initial integration of accelerometer code with the main control system.
Tested data stability and began debugging sensor responsiveness.

Week 7
Helped assemble the hardware module and ensured the accelerometer was properly mounted.
Collaborated with team members to refine sensor thresholds based on real-world motion tests.

Week 8

Helped with the installation and wiring of the LED strip onto the umbrella.
Conducted lighting tests to ensure proper functionality and visual effect.

Week 9
Assisted in final demo preparation, including system fine-tuning and stability tests.
Contributed to the project report writing, helped film the demo video, and recorded voiceover.

Week 10
Reviewed the full system to ensure all components worked correctly under real-use conditions.
Helped finalize the report formatting and presentation slides for submission.

Appendix E Code

1. NFC writing

```
 #include <Wire.h>
#include <Adafruit_PN532.h>

#define SDA_PIN A4
#define SCL_PIN A5

Adafruit_PN532 nfc(SDA_PIN, SCL_PIN);

void setup() {
   Serial.begin(9600);
   Wire.begin();
   nfc.begin();

   Serial.println("Scanning for PN532...");
   if (!nfc.getFirmwareVersion()) {
      Serial.println("PN532 not detected! Check wiring and I2C address");
      while (1);
   }

   nfc.SAMConfig();
   Serial.println("NFC Reader ready, waiting for card...");
}

void loop() {
   uint8_t uid[7];
   uint8_t uidLength;

   if (nfc.readPassiveTargetID(PN532_MIFARE_ISO14443A, uid, &uidLength)) {
      Serial.println("NFC card detected, writing data...");

      // Text to write
      char message[] = "Hello NFC!";
      if (nfc.ntag2xx_WritePage(4, (uint8_t*)message)) {
         Serial.println("Data written successfully!");
      } else {
         Serial.println("Write failed, check if card is writable!");
      }

      delay(5000);
   }
}
```

2. Acceleration and GPS module

```
#include <SoftwareSerial.h>
#include <TinyGPS++.h>
#include <Wire.h>
#include <MPU6050.h>

// 软串口定义
SoftwareSerial air780Serial(10, 11); // RX, TX (10接Air780EX的TX, 11接Air780EX的RX)
SoftwareSerial gpsSerial(7, 8);      // GPS模块 RX (7)，TX (8)

TinyGPSPlus gps;
MPU6050 mpu;

const int buzzerPin = 9;    // 蜂鸣器引脚
const int buttonPin = 6;    // 按键信号引脚（注意别用7, 7已连GPS）
bool detectionEnabled = false;  // 是否启用加速度检测
const float movementThreshold = 0.5;  // 加速度阈值

void setup() {
  Serial.begin(9600);
  air780Serial.begin(9600);
  gpsSerial.begin(9600);

  Wire.begin();
  mpu.initialize();

  pinMode(buzzerPin, OUTPUT);
  digitalWrite(buzzerPin, LOW);

  pinMode(buttonPin, INPUT);

  if (mpu.testConnection()) {
    Serial.println("MPU6050 连接成功");
  } else {
    Serial.println("MPU6050 连接失败，请检查接线");
    while (1);
  }

  Serial.println("按键控制加速度检测 (按下启用/关闭)");

  // 配置 GPS 模块只输出 RMC 数据
  gpsSerial.println("$PMTK314,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0*29");
  delay(1000);

  sendATCommand("AT+CGATT=1", 2000);
  sendATCommand("AT+CSTT", 2000);
  sendATCommand("AT+CIICR", 3000);
  sendATCommand("AT+CIFSR", 2000);

  Serial.println("Air780EX 已连接，开始获取 GPS...");
}

void loop() {
  handleButton();

  if (detectionEnabled) {
    detectMovement();
  }

  while (gpsSerial.available()) {
    char c = gpsSerial.read();
    gps.encode(c);
  }

  if (gps.location.isUpdated()) {
    float lat = gps.location.lat();
```

```
        float lon = gps.location.lng();
        Serial.print("lat: "); Serial.println(lat, 6);
        Serial.print("lon: "); Serial.println(lon, 6);

        gpsSerial.println("$PMTK314,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0*28");
        delay(3000);
        while (gpsSerial.available()) { gpsSerial.read(); }
        Serial.println("GPS 清空缓冲区");

        air780Serial.listen();
        while (air780Serial.available()) { air780Serial.read(); }

        String gpsData = "{\"lat\": " + String(lat, 6) + ", \"lon\": " + String(lon, 6) + "}";
        int len = gpsData.length();
        Serial.print("gpsData 长度: "); Serial.println(len);

        delay(5000);

        sendATCommand("AT+CSQ", 100);
        sendATCommand("AT+CGATT?", 200);
        sendATCommand("AT+CIFSR", 500);

        sendATCommand("AT+HTTPSSL=1", 100);
        sendATCommand("AT+HTTPSSL?", 100);
        sendATCommand("AT+HTTPINIT", 5000);
        sendATCommand("AT+HTTPPARA=\"URL\",\"https://gps-signal.glitch.me/gps\"", 4000);
        sendATCommand("AT+HTTPPARA=\"CONTENT\",\"application/json\"", 2000);

        delay(2000);
        checkURL();

        sendATCommand("AT+HTTPDATA=" + String(gpsData.length()) + ",20000", 5000);
        delay(5000);
        air780Serial.println(gpsData);

        sendATCommand("AT+HTTPACTION=1", 5000);
        delay(1000);
        sendATCommand("AT+HTTPREAD", 3000);
        sendATCommand("AT+HTTPTERM", 1000);

        gpsSerial.listen();
        gpsSerial.println("$PMTK314,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0*29");
        delay(500);
        delay(5000);
    }
}

void handleButton() {
    static bool lastButtonState = HIGH;
    bool buttonState = digitalRead(buttonPin);

    if (buttonState == LOW && lastButtonState == HIGH) {
        detectionEnabled = !detectionEnabled;
        Serial.println(detectionEnabled ? "加速度检测已启用" : "加速度检测已关闭");
        delay(200);
    }
    lastButtonState = buttonState;
}

void detectMovement() {
    int16_t ax, ay, az;
    mpu.getAcceleration(&ax, &ay, &az);

    float ax_g = ax / 16384.0;
    float ay_g = ay / 16384.0;
    float az_g = az / 16384.0;
    float totalAcc = sqrt(ax_g * ax_g + ay_g * ay_g + az_g * az_g);
```

```
    Serial.print("加速度: ");
    Serial.println(totalAcc);

    if (totalAcc > 1.0 + movementThreshold || totalAcc < 1.0 - movementThreshold) {
        Serial.println("警报！雨伞被移动！");
        digitalWrite(buzzerPin, HIGH);
        delay(2000);
        digitalWrite(buzzerPin, LOW);
    }
}

void sendATCommand(String cmd, int waitTime) {
    air780Serial.println(cmd);
    Serial.print("发送命令: "); Serial.println(cmd);

    unsigned long startTime = millis();
    while (millis() - startTime < waitTime) {
        while (air780Serial.available()) {
            char c = air780Serial.read();
            Serial.write(c);
        }
    }
}

void checkURL() {
    sendATCommand("AT+HTTPPARA?", 4000);
}
```

Appendix F Overview of pin connections

Module / Component	Arduino Uno Pin	Description
MPU6050 (Accelerometer & Gyroscope)	A4 (SDA)	I2C data line – connect to MPU6050's SDA
	A5 (SCL)	I2C clock line – connect to MPU6050's SCL
	VCC	3.3V or 5V (depends on your module's specification)
	GND	Ground
GT-U8 GPS Module	D7 (Software RX)	Connect to GPS module's TX pin
	D8 (Software TX)	(Optional) Connect to GPS RX – not always needed
	VCC	3.3V or 5V depending on your module
	GND	Ground
Air780E Module	D10 (Software RX)	Connect to Air780E TX pin
	D11 (Software TX)	Connect to Air780E RX pin
	VCC	Typically requires 3.8V – use a voltage regulator
	GND	Ground
Buzzer	D9	Digital output – HIGH to activate buzzer
	GND	Ground
Push Button	D6	Digital input – toggles motion detection mode